Security Test Report

MESample — Automotive 3D HMI Sample Project

Software Name	MESample
Software Version	1.3.0
Report Date	2026-09-09
Source Code Files	N/A (source code not delivered)
Delivered Artifacts	4 APK, 149.94 MB (HMI / SR / RenderService / UsedAsLib)
Report Scope	Static Code Analysis, Malware Scan, Fuzz Testing, Port Scan

1. Executive Summary

0
Security defects
found by static analysis

0
Malware threats
found in delivered APK

0
Fuzz crashes
in 1M runs

All 4 delivered APK passed malware scanning. Static code analysis reported **zero security vulnerabilities**. Fuzz testing completed 1,000,000 executions with **zero crashes**. Port scan confirmed **no unexpected network exposure**. All Category 5 security tests **PASSED**.

2. Test Summary

Test	Tool	Result	Detailed Report
Static Code Analysis	clang-tidy 21.0.0	**PASS (0 defects)**	01_clang-tidy/MESample_1.3.0_Static_Code_Analysis_Report.pdf
Virus Scan	Microsoft Defender	**PASS (4/4 APK clean)**	02_Defender/MESample_1.3.0_Virus_Scan_Report.pdf

Fuzz Testing	libFuzzer (LLVM 18)	**PASS (1M runs, 0 crashes)**	03_libFuzzer/MESample_1.3.0_Fuzz_Test_Report.pdf
Port Scan	Nmap 7.80	**PASS (no exposure)**	04_Nmap/MESample_1.3.0_Port_Scan_Report.pdf

3. Static Code Analysis

Category	Count	Severity
Security defects (clang-analyzer)	0	**Pass**
Potential bug patterns (bugprone)	17	**Low**
Performance suggestions	19	Info
Best-practice / style	416	Info
Total diagnostics	**452**	—

All findings are code-quality issues; none are security vulnerabilities. Compiled with `-Wall -Wextra -Werror`.

4. Malware Scan

#	Artifact	Size (MB)	Result
1	MESample_HMI_..._1920X1200_30fps.apk	9.18	**No threats found**
2	MESample_SR_..._1920X1080_30fps.apk	9.18	**No threats found**
3	MESample_RenderService_..._release.apk	69.50	**No threats found**
4	MESample_UsedAsLib_..._1920X1200_30fps.apk	62.07	**No threats found**

5. Fuzz Testing

Metric	Value	Status
Total executions	1,000,000	—
Crashes / hangs	0	**Pass**

Memory errors (ASan)	0	**Pass**
Invariant violations	0	**Pass**

6. Port Scan

PORT	STATE	SERVICE	Note
135/tcp	open	msrpc	Windows system default (RPC), unrelated
445/tcp	open	microsoft-ds	Windows system default (SMB), unrelated

Both open ports are Windows OS defaults; the software introduces no additional port exposure.

7. Conclusion

Item	Status
Static code analysis (security)	**PASS — no security defects**
Malware scan	**PASS — 4/4 APK clean**
Fuzz testing	**PASS — 1M runs, no crashes**
Port scan	**PASS — no unexpected exposure**

- No security vulnerabilities identified in the self-developed C++ source code.
- All 4 delivered APK artifacts are free of malware.
- Fuzz testing of input-parsing logic found no crashes in 1M executions.
- No unexpected network port exposure introduced by the software.